

- 4 Fig. 4.1 is a scanning electron micrograph showing the tissue that lines the bronchi in the gas exchange system.

Fig. 4.2 is a transmission electron micrograph of a horizontal section made at the position indicated by the two arrows in Fig. 4.1.

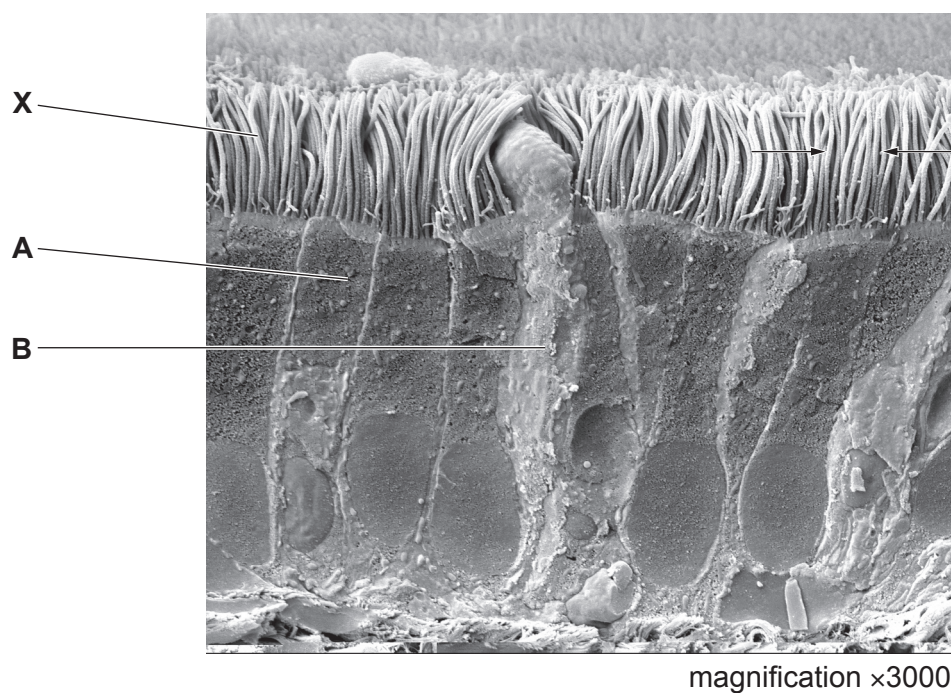


Fig. 4.1

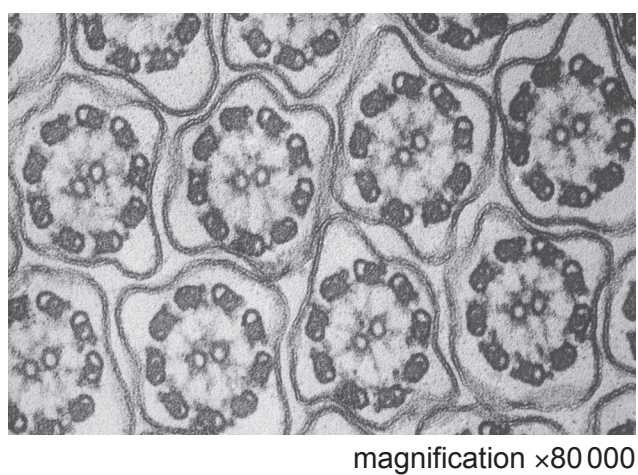


Fig. 4.2

- (a) (i) Name the cells labelled **A** and **B** in Fig. 4.1.

A

B

- (ii) Describe how the tissue shown in Fig. 4.1 is adapted to its function in the gas exchange system.

.....

.....

.....

.....

.....

.....

..... [3]

- (b) (i) The structures labelled **X** in Fig. 4.1 have a characteristic internal appearance, as seen in Fig. 4.2.

Describe the internal appearance of the structures labelled **X**.

.....

.....

.....

.....

..... [2]

- (ii) Explain how Fig. 4.2 shows that each of the structures labelled **X** are intracellular.

.....

..... [1]

(c) Stem cells are found in the lining of the bronchi.

Describe the function of centrioles **and** explain how they are involved in the cell cycle of a stem cell.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 12]